## Supplementary material

Channel attention in lightweight hyperspectral CNNs: an equivalence-tested comparison

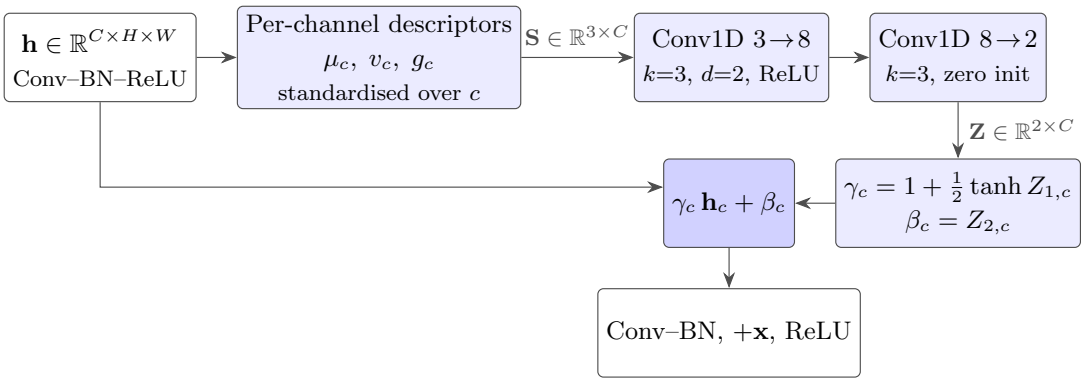



Figure S1: The SNSM probe inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ (mean, variance, cross-channel gradient) are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale $\gamma_c$ and a shift $\beta_c$ for each channel. The second convolution starts at zero, so the block is initially the identity.

Table S1: Random 5% splits: OA and AA (%, mean ± std) of all models, including an FC ($\gamma$+$\beta$) gate and two external baselines trained under the same protocol, HybridSN (original layer configuration) and SSFTT (four tokens, one transformer layer with eight heads). The eight backbone variants use 15 seeds on IP and five elsewhere; the other models use five.

| Method | IP | | KSC | | PU | | BW | |
|---|---|---|---|---|---|---|---|---|
| | OA | AA | OA | AA | OA | AA | OA | AA |
| Plain CNN | 96.28±0.79 | 89.49±2.44 | 99.65±0.34 | 99.51±0.36 | **99.59±0.07** | 99.01±0.16 | 99.09±0.55 | 98.87±0.90 |
| SE | 96.34±0.62 | 90.04±2.18 | 99.61±0.28 | 99.43±0.38 | 99.53±0.19 | 98.80±0.56 | 98.95±0.82 | 98.76±1.07 |
| ECA | 96.28±0.68 | 90.87±2.53 | 99.61±0.47 | 99.40±0.66 | 99.58±0.10 | 99.00±0.20 | **99.13±0.76** | **98.90±1.08** |
| CBAM | 96.19±0.60 | 91.58±1.49 | 99.54±0.56 | 99.23±0.83 | 99.58±0.06 | **99.11±0.18** | 98.99±0.87 | 98.78±1.20 |
| GCT | 96.28±0.80 | 90.07±2.34 | **99.69±0.26** | **99.55±0.28** | 99.58±0.08 | 98.99±0.18 | 99.11±0.56 | 98.88±0.91 |
| SRM | 96.20±0.71 | 93.04±2.14 | 99.69±0.36 | 99.51±0.49 | 99.55±0.07 | 98.92±0.17 | 98.87±0.63 | 98.67±0.97 |
| SNSM ($\gamma$) | 96.17±0.73 | 92.89±2.00 | 99.69±0.26 | 99.50±0.39 | 99.49±0.05 | 98.91±0.16 | 99.10±0.62 | 98.89±0.96 |
| SNSM ($\gamma$+$\beta$) | 96.27±0.67 | **93.33±1.68** | 99.63±0.55 | 99.45±0.73 | 99.47±0.05 | 98.93±0.16 | 99.04±0.69 | 98.85±1.03 |
| FC ($\gamma$+$\beta$) | 96.24±0.52 | 92.51±1.60 | 99.63±0.40 | 99.48±0.47 | 99.51±0.06 | 98.88±0.14 | 98.80±0.70 | 98.63±0.98 |
| HybridSN | 88.30±4.91 | 73.64±8.28 | 95.00±3.02 | 91.20±5.32 | 97.84±1.52 | 95.46±4.12 | 77.16±37.92 | 75.56±38.31 |
| SSFTT | **96.71±0.79** | 92.35±1.81 | 99.38±0.28 | 99.15±0.32 | – | – | – | – |

Table S2: Paired AA differences SNSM ($\gamma$+$\beta$) − block (points) with 95% $t$ intervals, for every setting of the main paper's paired table.

| Setting | Plain | SE | ECA | CBAM | GCT | SRM | SNSM ($\gamma$) | Seeds |
|---|---|---|---|---|---|---|---|---|
| IP, random 5% | +3.84 [+2.4, +5.3] | +3.29 [+2.2, +4.4] | +2.45 [+0.9, +4.0] | +1.75 [+0.9, +2.6] | +3.26 [+2.0, +4.6] | +0.29 [-0.8, +1.4] | +0.44 [-0.3, +1.2] | 15 |
| IP, smallest 15 | +0.39 [+0.1, +0.7] | +0.50 [+0.3, +0.7] | +0.33 [+0.1, +0.6] | +0.60 [+0.3, +0.9] | +0.35 [+0.0, +0.7] | +0.41 [+0.0, +0.8] | +0.30 [+0.1, +0.5] | 15 |
| IP, random 50/15 | -0.01 [-0.2, +0.2] | -0.08 [-0.2, +0.1] | +0.04 [-0.1, +0.2] | +0.05 [-0.1, +0.2] | +0.01 [-0.2, +0.2] | +0.07 [-0.1, +0.2] | +0.06 [-0.2, +0.3] | 15 |
| IP, standard | +0.24 [+0.0, +0.4] | +0.23 [+0.0, +0.4] | +0.33 [+0.2, +0.5] | +0.04 [-0.2, +0.2] | +0.25 [+0.1, +0.4] | -0.02 [-0.2, +0.2] | +0.12 [-0.1, +0.3] | 15 |
| PU, smallest 1/1/2 | -0.59 [-1.6, +0.4] | -0.74 [-1.6, +0.1] | -1.19 [-1.9, -0.5] | -0.30 [-1.1, +0.5] | -1.14 [-2.2, -0.1] | -0.55 [-1.2, +0.2] | -1.59 [-2.6, -0.5] | 15 |
| KSC, smallest 1/1/2 | -0.24 [-0.9, +0.5] | +0.79 [+0.1, +1.4] | +0.39 [-0.6, +1.4] | +0.54 [-0.7, +1.8] | -0.21 [-1.0, +0.5] | +1.41 [+0.3, +2.5] | +0.03 [-1.3, +1.3] | 15 |
| PU, hardest 1/1/2 | +0.97 [-0.1, +2.1] | +0.98 [-0.2, +2.1] | +0.64 [-0.5, +1.8] | +0.96 [-0.4, +2.3] | +0.77 [-0.2, +1.8] | +0.79 [-0.3, +1.9] | -0.82 [-1.5, -0.1] | 15 |
| KSC, hardest 1/1/2 | +0.71 [-0.0, +1.4] | +0.88 [-0.1, +1.8] | +0.20 [-0.5, +0.9] | +1.14 [+0.3, +2.0] | +0.77 [+0.1, +1.4] | +0.66 [-0.3, +1.6] | +0.90 [-0.3, +2.1] | 15 |
| WHU, random 5% | -0.06 [-0.1, +0.0] | -0.01 [-0.1, +0.1] | -0.02 [-0.1, +0.0] | +0.00 [-0.1, +0.1] | +0.00 [-0.1, +0.1] | -0.06 [-0.2, +0.0] | -0.03 [-0.1, +0.1] | 10 |
| PU, patch-disjoint | +0.31 [-1.9, +2.5] | -0.47 [-2.7, +1.8] | -0.39 [-2.4, +1.7] | +1.03 [-0.7, +2.8] | +0.92 [-1.0, +2.8] | +1.33 [-0.4, +3.1] | +0.03 [-2.0, +2.0] | 15 |
| KSC, patch-disjoint | +3.46 [+1.5, +5.4] | -0.19 [-2.3, +1.9] | +1.86 [+0.3, +3.4] | +0.91 [-0.9, +2.7] | +3.51 [+1.5, +5.5] | +4.25 [+1.7, +6.8] | +1.40 [-1.0, +3.8] | 15 |
| WHU, patch-disjoint | -2.01 [-3.3, -0.7] | -2.39 [-4.8, +0.1] | -1.77 [-4.5, +1.0] | -1.85 [-3.6, -0.1] | -1.32 [-3.3, +0.6] | -0.84 [-2.7, +1.0] | -0.46 [-3.0, +2.1] | 10 |
| PU, block-disjoint | -3.77 [-5.6, -2.0] | -2.24 [-3.9, -0.5] | -2.95 [-4.7, -1.2] | +0.72 [-1.1, +2.5] | -4.04 [-5.7, -2.4] | -1.14 [-2.6, +0.3] | -1.16 [-2.7, +0.4] | 15 |
| KSC, block-disjoint | -0.63 [-3.4, +2.1] | -1.66 [-4.6, +1.3] | -0.99 [-3.6, +1.6] | -0.56 [-2.3, +1.2] | -0.56 [-3.5, +2.3] | +1.64 [-1.3, +4.6] | -1.24 [-3.6, +1.1] | 15 |
| IP, random 5%, deeper host | -0.53 [-1.4, +0.3] | +0.14 [-0.6, +0.9] | +0.18 [-0.5, +0.9] | -0.35 [-1.2, +0.5] | -0.51 [-1.3, +0.2] | -0.87 [-2.2, +0.5] | -0.19 [-1.3, +0.9] | 10 |
| KSC, patch-disjoint, deeper host | -1.88 [-4.9, +1.1] | -0.80 [-5.5, +3.9] | -2.87 [-5.2, -0.6] | -1.35 [-4.7, +2.0] | -1.63 [-4.6, +1.3] | +0.34 [-2.8, +3.4] | -2.00 [-5.6, +1.6] | 10 |
| KSC, patch-disjoint, train-only PCA | +0.33 [-2.3, +2.9] | -0.98 [-3.1, +1.2] | -0.01 [-2.4, +2.4] | -2.38 [-4.1, -0.7] | +0.54 [-2.0, +3.1] | +1.65 [-0.7, +4.0] | -0.38 [-2.8, +2.1] | 15 |
| IP, random 5%, lr $3 \times 10^{-4}$ | +8.16 [+6.8, +9.5] | +7.42 [+4.8, +10.0] | +8.26 [+5.4, +11.2] | +6.12 [+3.5, +8.8] | +8.06 [+6.7, +9.4] | +5.37 [+4.1, +6.6] | +5.27 [+3.1, +7.4] | 5 |
| IP, random 5%, lr $3 \times 10^{-3}$ | +0.98 [-0.0, +2.0] | +1.48 [+0.2, +2.8] | +0.56 [-0.3, +1.4] | +0.80 [-0.3, +1.9] | +0.83 [+0.0, +1.7] | +0.28 [-0.8, +1.3] | -0.03 [-1.2, +1.2] | 15 |

Table S3: Paired AA differences block − plain CNN (points), so that no block is the reference. *: Holm-corrected Wilcoxon $p < 0.05$ within the row (seven comparisons); †: equivalent within ±1 point (paired TOST). GCT stays close to the plain CNN in most settings because its gate starts at the identity.

| Setting | SE | ECA | CBAM | GCT | SRM | SNSM ($\gamma$) | SNSM ($\gamma$+$\beta$) |
|---|---|---|---|---|---|---|---|
| IP, random 5% | +0.55 | +1.38 | +2.09* | +0.58*† | +3.55* | +3.40* | +3.84* |
| IP, smallest 15 | -0.11† | +0.06† | -0.20† | +0.04† | -0.02† | +0.09† | +0.39*† |
| IP, random 50/15 | +0.07† | -0.05† | -0.06† | -0.01† | -0.08† | -0.07† | -0.01† |
| IP, standard | +0.01† | -0.09† | +0.21† | -0.01† | +0.26*† | +0.13† | +0.24† |
| PU, smallest 1/1/2 | +0.15† | +0.60 | -0.29 | +0.55† | -0.05† | +0.99 | -0.59 |
| KSC, smallest 1/1/2 | -1.03* | -0.63 | -0.78 | -0.03† | -1.65* | -0.27 | -0.24† |
| PU, hardest 1/1/2 | -0.01† | +0.33 | +0.01† | +0.20† | +0.18† | +1.79* | +0.97 |
| KSC, hardest 1/1/2 | -0.17† | +0.52 | -0.43 | -0.06† | +0.06† | -0.19 | +0.71 |
| WHU, random 5% | -0.05† | -0.03† | -0.06† | -0.06† | +0.00† | -0.03† | -0.06† |
| PU, patch-disjoint | +0.78 | +0.70 | -0.72 | -0.61 | -1.02 | +0.28 | +0.31 |
| KSC, patch-disjoint | +3.65* | +1.60 | +2.55 | -0.06† | -0.79 | +2.06 | +3.46* |
| WHU, patch-disjoint | +0.37 | -0.24 | -0.16 | -0.69 | -1.17 | -1.55 | -2.01 |
| PU, block-disjoint | -1.53 | -0.82 | -4.49* | +0.27† | -2.63* | -2.61* | -3.77* |
| KSC, block-disjoint | +1.04 | +0.36 | -0.07 | -0.06† | -2.27 | +0.61 | -0.63 |
| IP, random 5%, deeper host | -0.67 | -0.70 | -0.18† | -0.01† | +0.34 | -0.33 | -0.53 |
| KSC, patch-disjoint, deeper host | -1.08 | +0.98 | -0.54 | -0.25 | -2.22 | +0.12 | -1.88 |
| KSC, patch-disjoint, train-only PCA | +1.31 | +0.35 | +2.71 | -0.20† | -1.32 | +0.72 | +0.33 |
| IP, random 5%, lr $3 \times 10^{-4}$ | +0.74 | -0.10 | +2.04 | +0.10† | +2.79 | +2.89 | +8.16 |
| IP, random 5%, lr $3 \times 10^{-3}$ | -0.50 | +0.42 | +0.18 | +0.14† | +0.70 | +1.01 | +0.98 |

Table S4: Sensitivity to the equivalence margin: number of the seven comparisons SNSM ($\gamma$+$\beta$) − block per setting that are equivalent (paired TOST, $\alpha = 0.05$) within ±0.5, ±1 (main paper) and ±2 AA points.

| Setting | ±0.5 | ±1 | ±2 |
|---|---|---|---|
| IP, random 5% | 0 | 0 | 2 |
| IP, smallest 15 | 1 | 7 | 7 |
| IP, random 50/15 | 7 | 7 | 7 |
| IP, standard | 7 | 7 | 7 |
| PU, smallest 1/1/2 | 0 | 1 | 5 |
| KSC, smallest 1/1/2 | 0 | 2 | 6 |
| PU, hardest 1/1/2 | 0 | 0 | 6 |
| KSC, hardest 1/1/2 | 0 | 1 | 7 |
| WHU, random 5% | 7 | 7 | 7 |
| PU, patch-disjoint | 0 | 0 | 1 |
| KSC, patch-disjoint | 0 | 0 | 1 |
| WHU, patch-disjoint | 0 | 0 | 0 |
| PU, block-disjoint | 0 | 0 | 0 |
| KSC, block-disjoint | 0 | 0 | 1 |
| IP, random 5%, deeper host | 0 | 2 | 7 |
| KSC, patch-disjoint, deeper host | 0 | 0 | 0 |
| KSC, patch-disjoint, train-only PCA | 0 | 0 | 1 |
| IP, random 5%, lr $3 \times 10^{-4}$ | 0 | 0 | 0 |
| IP, random 5%, lr $3 \times 10^{-3}$ | 0 | 0 | 6 |

Table S5: Paired OA differences SNSM ($\gamma$+$\beta$) − block (points). *: Holm-corrected Wilcoxon $p < 0.05$ within the row; †: equivalent within ±1 point (paired TOST).

| Setting | Plain | SE | ECA | CBAM | GCT | SRM | SNSM ($\gamma$) | Seeds |
|---|---|---|---|---|---|---|---|---|
| IP, random 5% | -0.01† | -0.06† | -0.01† | +0.08† | -0.00† | +0.07† | +0.11† | 15 |
| IP, smallest 15 | -0.14† | -0.12† | -0.11† | +0.04† | -0.12† | +0.02† | +0.09† | 15 |
| IP, random 50/15 | +0.04† | -0.15† | +0.09† | +0.12† | +0.06† | +0.10† | +0.01† | 15 |
| IP, standard | +0.08† | +0.34† | +0.31† | +0.14† | +0.15† | +0.03† | +0.16† | 15 |
| PU, smallest 1/1/2 | -0.25† | -0.26† | -0.39*† | -0.10† | -0.39† | -0.18† | -0.47*† | 15 |
| KSC, smallest 1/1/2 | -0.20† | +0.23† | +0.17† | +0.19† | -0.19† | +0.50† | -0.03† | 15 |
| PU, hardest 1/1/2 | +0.53 | +0.58 | +0.31† | +0.50 | +0.40† | +0.50 | -0.55† | 15 |
| KSC, hardest 1/1/2 | +0.75 | +0.68* | +0.29† | +0.95* | +0.80 | +0.56 | +0.43 | 15 |
| WHU, random 5% | -0.04† | -0.01† | -0.01† | -0.02† | -0.03† | -0.05† | -0.02† | 10 |
| PU, patch-disjoint | +1.50 | +0.89 | +0.51 | +0.69 | +2.21 | +0.79 | +1.39 | 15 |
| KSC, patch-disjoint | +1.98 | +0.32 | +0.72 | +0.43 | +1.91 | +2.75 | +0.25 | 15 |
| WHU, patch-disjoint | -2.51 | -2.86 | -1.22 | -1.22 | -0.24 | -2.61 | +0.07 | 10 |
| PU, block-disjoint | -1.65 | -0.86 | -1.09 | -0.41 | -1.80 | -1.25 | -0.79 | 15 |
| KSC, block-disjoint | -0.27 | -1.57 | -0.92 | -0.04 | -0.19 | +0.77 | -0.80 | 15 |
| IP, random 5%, deeper host | -0.41† | -0.26† | -0.26† | -0.34† | -0.39† | -0.27† | -0.18† | 10 |
| KSC, patch-disjoint, deeper host | -1.12 | -0.79 | -2.40 | -1.66 | -1.23 | +1.66 | -2.13 | 10 |
| KSC, patch-disjoint, train-only PCA | -0.74 | -0.65 | +0.23 | -2.17 | -0.67 | -0.22 | -0.78 | 15 |
| IP, random 5%, lr $3 \times 10^{-4}$ | +0.45† | +0.29† | +0.29† | -0.06† | +0.46† | +0.27† | +0.18† | 5 |
| IP, random 5%, lr $3 \times 10^{-3}$ | -0.23† | -0.03† | -0.23† | -0.26† | -0.22† | -0.12† | -0.07† | 15 |

Table S6: AA (%, mean ± std) of random 5% splits and of the same splits with only the three smallest classes changed (15 seeds; the unchanged PU and KSC splits use five). The last row is the range of the block means.

| Block | IP, 5% | IP, smallest 15 | PU, 5% | PU, smallest 1/1/2 | KSC, 5% | KSC, smallest 1/1/2 |
|---|---|---|---|---|---|---|
| Plain CNN | 89.49±2.44 | 96.03±0.76 | 99.01±0.16 | 84.20±3.43 | 99.51±0.36 | **93.12±3.44** |
| SE | 90.04±2.18 | 95.93±0.64 | 98.80±0.56 | 84.35±3.29 | 99.43±0.38 | 92.09±3.25 |
| ECA | 90.87±2.53 | 96.10±0.76 | 99.00±0.20 | 84.80±3.23 | 99.40±0.66 | 92.48±3.19 |
| CBAM | 91.58±1.49 | 95.83±0.67 | **99.11±0.18** | 83.91±2.11 | 99.23±0.83 | 92.34±4.15 |
| GCT | 90.07±2.34 | 96.07±0.75 | 98.99±0.18 | 84.74±3.75 | **99.55±0.28** | 93.09±3.24 |
| SRM | 93.04±2.14 | 96.01±0.81 | 98.92±0.17 | 84.15±3.45 | 99.51±0.49 | 91.47±3.62 |
| SNSM ($\gamma$) | 92.89±2.00 | 96.13±0.72 | 98.91±0.16 | **85.19±3.48** | 99.50±0.39 | 92.84±3.64 |
| SNSM ($\gamma$+$\beta$) | **93.33±1.68** | **96.43±0.66** | 98.93±0.16 | 83.60±2.96 | 99.45±0.73 | 92.88±3.12 |
| Range of block means | 3.84 | 0.60 | 0.31 | 1.59 | 0.32 | 1.65 |

Table S7: Robustness checks: AA (%, mean ± std) of the eight backbone variants on block-disjoint splits of PU and KSC (15 seeds), in a deeper, wider host (four residual blocks of 128 channels; ten seeds), with PCA fitted on training pixels only (15 seeds), with the three hardest classes reduced to 1, 1 and 2 training pixels (15 seeds), and with learning rates $3 \times 10^{-4}$ (five seeds) and $3 \times 10^{-3}$ (15 seeds). The last row is the range of the block means.

| Block | PU, block-disj. | KSC, block-disj. | IP, deeper | KSC disj., deeper | KSC disj., train PCA | PU, hardest | KSC, hardest | IP, lr 3e-4 | IP, lr 3e-3 |
|---|---|---|---|---|---|---|---|---|---|
| Plain CNN | 83.20±4.34 | 71.45±6.96 | 93.85±1.69 | 60.46±5.77 | 64.26±4.88 | 73.19±2.03 | 88.17±1.35 | 80.23±1.51 | 92.15±2.12 |
| SE | 81.67±4.89 | **72.48±7.42** | 93.18±2.02 | 59.38±6.05 | 65.57±5.03 | 73.17±2.09 | 88.01±1.39 | 80.97±1.99 | 91.65±1.84 |
| ECA | 82.38±4.21 | 71.81±8.11 | 93.14±1.80 | **61.44±6.66** | 64.61±4.56 | 73.52±2.28 | 88.69±1.92 | 80.13±2.35 | 92.57±1.93 |
| CBAM | 78.71±4.62 | 71.38±6.95 | 93.67±1.95 | 59.92±6.28 | **66.97±6.19** | 73.20±2.22 | 87.74±2.14 | 82.27±1.98 | 92.33±1.75 |
| GCT | **83.46±4.31** | 71.38±6.93 | 93.83±1.61 | 60.21±6.45 | 64.06±5.14 | 73.39±2.05 | 88.11±1.45 | 80.33±1.43 | 92.30±2.04 |
| SRM | 80.57±4.08 | 69.18±7.01 | **94.19±1.91** | 58.24±5.28 | 62.94±5.31 | 73.37±2.56 | 88.23±2.26 | 83.02±1.64 | 92.85±1.75 |
| SNSM ($\gamma$) | 80.59±4.79 | 72.06±6.56 | 93.51±1.59 | 60.58±5.41 | 64.98±7.02 | **74.98±2.65** | 87.98±2.44 | 83.12±1.49 | **93.16±1.89** |
| SNSM ($\gamma$+$\beta$) | 79.43±5.06 | 70.82±7.87 | 93.32±2.21 | 58.58±7.23 | 64.59±6.63 | 74.16±2.58 | **88.89±1.34** | **88.39±0.78** | 93.13±2.17 |
| Range of block means | 4.76 | 3.30 | 1.05 | 3.20 | 4.03 | 1.81 | 1.14 | 8.26 | 1.51 |

Table S8: SNSM ($\gamma$+$\beta$) probe ablation on IP, random 5% splits (AA, %; five seeds). $\Delta$AA is the paired difference from the full block; wins count seeds on which the variant is better.

| Descriptors | AA | $\Delta$AA (wins/5) |
|---|---|---|
| mean, var, grad (SNSM) | 94.16±1.07 | – |
| mean | 93.01±0.72 | -1.16 (1/5) |
| mean, var | 94.55±1.67 | +0.39 (4/5) |
| mean, grad | 93.37±1.09 | -0.80 (1/5) |
| mean, var, grad; default init. | 92.67±1.15 | -1.49 (0/5) |

Table S9: Blocks applied to the raw bands in wavelength order, without PCA or wavelet features: AA (%, mean ± std, random 5% splits, five seeds). CBAM and SRM were not run in this setting.

| Block | IP | PU | KSC |
|---|---|---|---|
| Plain CNN | 82.53±0.95 | 99.00±0.31 | 99.18±0.43 |
| SE | 83.54±2.01 | 98.91±0.36 | 98.61±0.89 |
| ECA | 82.95±0.71 | **99.10±0.22** | **99.28±0.30** |
| GCT | 82.64±0.87 | 99.07±0.23 | 99.18±0.60 |
| SNSM ($\gamma$) | 83.56±1.08 | 99.10±0.21 | 98.96±0.86 |
| SNSM ($\gamma$+$\beta$) | **87.45±2.48** | 98.87±0.25 | 98.83±0.89 |

Table S10: Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random 5% splits). MMACs: multiply–accumulates of convolution and linear layers counted by `thop` (attention products and descriptor statistics excluded). Latency: batch size 1 on a Tesla T4.

| Method | Params (M) | MMACs | Latency (ms) | Train IP (s) | Train PU (s) |
|---|---|---|---|---|---|
| Plain CNN | 0.184 | 41.2 | 0.69 | 4.2 | 17.6 |
| SE | 0.188 | 41.2 | 1.17 | 4.5 | 19.3 |
| ECA | 0.184 | 41.2 | 0.98 | 4.8 | 19.8 |
| CBAM | 0.185 | 41.3 | 1.55 | 5.9 | 24.9 |
| GCT | 0.184 | 41.2 | 1.11 | 4.7 | 20.0 |
| SRM | 0.184 | 41.2 | 1.22 | 4.8 | 20.2 |
| SNSM ($\gamma$) | 0.184 | 41.3 | 2.54 | 7.1 | 29.5 |
| SNSM ($\gamma$+$\beta$) | 0.184 | 41.3 | 2.54 | 7.2 | 30.2 |
| FC ($\gamma$+$\beta$) | 0.190 | 41.2 | 1.31 | 4.7 | 19.8 |
| HybridSN | 1.743 | 137.4 | 0.71 | 26.2 | 115.3 |
| SSFTT | 0.291 | 34.9 | 1.25 | 7.6 | – |